

DILLON 18 WSW, MT, USA

WMO: 742270

Lat: 45.1581N

Lon: 113.0057W

Elev: 5971

StdP: 11.79

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 04137

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-17.3	-10.0	-21.9	2.1	-16.2	-15.0	3.0	-9.5	19.3	27.1	17.1	27.8	1.9	N/A	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.9	86.2	53.9	83.9	53.6	81.0	53.0	57.2	77.5	55.8	76.7	54.6	75.8	9.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
50.1	67.1	61.2	47.7	61.1	60.2	45.7	56.6	59.3	27.1	77.3	26.1	76.7	25.2	75.9	64.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.3	14.2	DB	-28.5	90.9	6.7	1.5	-33.3	92.0	-37.2	92.9	-41.0	93.8	-45.8	94.9	(4)
(5)			WB	-28.4	60.4	7.7	2.2	-33.9	62.0	-38.4	63.3	-42.7	64.5	-48.2	66.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.0	18.7	20.4	30.0	37.4	46.1	54.7	63.1	60.8	52.2	39.8	27.2	16.3		
	DBStd	18.11	11.37	11.48	9.32	7.16	6.87	6.73	4.39	4.71	7.06	8.32	10.91	11.59		
	HDD50	5106	970	830	620	381	158	32	0	2	57	326	685	1045		
	HDD65	9531	1435	1250	1085	828	586	312	86	140	384	780	1135	1510		
	CDD50	1090	0	0	0	3	37	173	405	338	124	10	0	0		
	CDD65	42	0	0	0	0	0	4	26	11	1	0	0	0		
	CDH74	2854	0	0	0	0	18	306	1310	970	241	9	0	0		
	CDH80	752	0	0	0	0	0	71	385	257	39	0	0	0		
	WSAvg	5.1	4.6	5.1	5.3	6.4	5.6	5.5	5.1	4.8	4.7	5.0	4.5	4.2		
	PrecAvg	11.2	0.6	0.4	0.6	1.1	1.6	2.0	1.0	0.9	0.8	0.9	0.7	0.6		
	PrecMax	17.0	1.4	1.0	1.5	1.8	2.7	3.4	3.5	1.9	1.7	2.2	2.0	1.1		
(15) Precipitation	PrecMin	8.0	0.1	0.0	0.2	0.3	0.7	0.6	0.0	0.2	0.0	0.0	0.1	0.2		
	PrecStd	2.4	0.4	0.2	0.3	0.4	0.6	0.9	0.8	0.6	0.5	0.6	0.5	0.3		
	DB	44.4	48.4	60.4	68.8	76.6	86.2	88.9	89.0	84.9	73.6	58.2	44.5			
	MCWB	34.7	36.6	41.4	45.5	49.3	53.7	55.2	53.8	52.7	50.1	43.0	35.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	40.1	42.4	53.0	64.0	72.2	81.8	86.5	85.9	79.9	65.8	52.6	40.2			
	2% MCWB	33.3	34.8	39.1	44.0	48.7	53.7	54.2	53.6	51.1	46.0	40.4	33.8			
	5% DB	37.4	38.8	48.0	58.8	67.6	77.3	84.6	83.1	76.6	61.4	47.6	36.6			
	5% MCWB	31.5	32.6	37.2	41.6	47.5	52.4	54.0	52.8	50.2	44.6	37.6	31.3			
	10% DB	34.1	36.1	43.9	53.0	62.7	72.9	81.9	79.9	72.5	56.0	43.3	32.9			
	10% MCWB	29.3	30.8	35.2	39.2	45.8	51.2	53.7	52.2	49.3	42.4	35.2	28.3			
	0.4% WB	36.8	37.8	43.7	48.0	52.8	58.6	59.9	58.0	57.1	50.7	44.1	36.9			
	0.4% MCDB	42.5	46.9	57.0	64.7	68.0	78.7	80.1	78.7	69.7	69.4	57.1	42.6			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	34.2	35.5	40.6	44.7	50.4	56.0	57.5	56.0	53.7	48.3	41.0	34.8			
	2% MCDB	39.9	42.1	52.2	61.4	65.7	77.2	77.6	76.6	72.8	63.0	51.0	39.7			
	5% WB	32.2	33.5	38.5	42.4	48.5	53.9	56.3	54.7	51.8	45.9	38.4	31.8			
	5% MCDB	37.6	39.2	47.8	57.2	63.8	72.7	77.5	76.1	72.4	58.8	47.3	36.5			
	10% WB	30.0	31.6	36.2	40.2	46.9	52.3	55.1	53.5	50.2	43.5	35.2	28.7			
	10% MCDB	34.7	36.8	44.5	52.5	61.1	70.7	77.0	75.9	69.9	55.8	42.2	32.9			
	MDBR	20.6	20.0	20.8	23.4	24.5	28.2	34.9	35.2	31.9	25.4	22.8	20.8			
	5% DB	20.9	17.8	26.5	35.3	35.5	38.2	39.8	41.5	41.9	36.2	29.3	20.4			
(36) Mean Daily Temperature Range	MCWBR	15.5	12.4	14.5	18.0	16.1	15.5	14.6	16.1	18.3	18.5	18.2	15.2			
	MCDBR	20.0	17.4	24.7	32.4	29.5	34.2	35.5	35.8	36.3	31.9	27.7	19.2			
	MCWBR	14.8	12.2	14.1	16.8	13.9	14.2	13.8	14.6	16.2	16.7	17.3	14.6			
	MCDBR	14.8	12.2	14.1	16.8	13.9	14.2	13.8	14.6	16.2	16.7	17.3	14.6			
(40) Clear-Sky Solar Irradiance	taub	0.229	0.238	0.255	0.292	0.300	0.306	0.312	0.316	0.292	0.268	0.246	0.229			
	taud	2.417	2.391	2.476	2.447	2.514	2.534	2.541	2.491	2.560	2.598	2.563	2.474			
	Ebn at Noon	291	307	315	308	307	305	301	297	298	293	284	281			
	Edh at Noon	22	28	30	33	32	32	31	32	27	23	19	19			
(44) All-Sky Solar Radiation	RadAvg	520	832	1243	1613	1846	2142	2266	1954	1496	962	585	423			
	RadStd	50	72	78	116	174	170	122	108	123	75	46	33			

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-1.19	N/A	N/A	N/A	N/A

Nomenclature: See separate page